Laser Beam Propagation Through Air

I created a simulation of the irradiance profile of a laser beam focused at a specific distance.

In this artificial simulation, atmospheric turbulence—primarily caused by random variations in the refractive index of air—increases over a period of ten seconds, with the turbulence parameter (C_n^2) rising from (10^{-17}) to (10^{-13},\mathrm{m^{-2/3}}).

In real atmospheric conditions, fluctuations can occur on much shorter timescales and over a considerably wider range of spatial and temporal scales but with less variance of C_n^2. Nevertheless, this simulation illustrates how increasing turbulence causes the laser beam to wander randomly and its diameter to increase, eventually producing pronounced beam spreading and intensity fluctuations.

Technically, the beam quality could be corrected using adaptive optics or a phased array.